

- **LLM Inference Engine**

A from-scratch LLM inference engine in PyTorch (KV cache, batching, paged memory)

[Github](#)

- Built a PyTorch LLM engine for GPT-2/Mistral with KV caching, continuous batching, paged memory, and async APIs.
- Increased decode throughput 2.22x and cut CUDA execution time 46.6% using a preallocated KV cache on one A100 GPU.
- Scaled GPT-2 124M throughput from 155 to 18,346 tokens/second, a 118.3x gain at batch size 512 on one A100 GPU.
- Cut KV-cache memory 35.9%, reached 3.2x throughput at 64 concurrent requests, and validated behavior with 122 tests.